

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

def high_boost_filter(image, k):
    # Convert to float for precise calculations
    img_float = image.astype(np.float32)

    # 1. Blur the image using a Gaussian filter
    blurred = cv2.GaussianBlur(img_float, (3, 3), 0)

    # 2. Create the mask (Unsharp Mask): Original - Blurred
    mask = img_float - blurred

    # 3. High-Boost filtering: Original + k * Mask
    # If k=1, it is unsharp masking. If k > 1, it is high-boost filtering.
    boosted = img_float + k * mask

    # Clip and convert back to uint8
    return np.clip(boosted, 0, 255).astype(np.uint8)

# Load the image from the Colab environment
image_path = "image1.jpg"
image = cv2.imread(image_path)

if image is not None:
    # Apply high-boost filter with a boost factor (e.g., k=1.5)
    sharpened_image = high_boost_filter(image, 1.5)

    # Display and Save
    print("Original Image:")
    cv2_imshow(image)
    print("\nSharpened Image (High-Boost Filtering, k=1.5):")
    cv2_imshow(sharpened_image)

    cv2.imwrite('high_boost_sharpened.jpg', sharpened_image)
else:
    print(f"Error: '{image_path}' not found. Please ensure the image is up.
```

Original Image:



Sharpened Image (High-Boost Filtering, $k=1.5$):

